SEP 2017 DATA STURCTURES IN C 80 MARKS

SECTION -A

1). ANSWER ALL QUESTIONS. **10*2=20**

a) Define data structures, list out its types.
b) Define pointer? How to initialize pointers.
c) Explain fprintf() , fscanf(), fopen().
d) List out primitive and non-primitive data structures.
e) Define linear search.
f) What is recursion? Give example.
g) Define stack? List out the operations of stack.
h) Convert the given expression from infix to postfix
   1) A/B*(C+D/E)-F
i) What is circular queue.
j) Define doubly linked list and list out its operations.

SECTION – B

Answer any four questions **4*5=20**

2) Explain with syntax malloc(), calloc(), free(), and realloc().
3) Write a c program for insertion sort.
4) Write a code for insert an element in stack.
5) Differentiate between Queue and double ended queue.
6) What are the advantages and disadvantages of linked lists

SECTION – C

Answer any four questions **4*10=40**

7) Write a c program to store student name, Reg no, date of birth, class of a two students, using file.
8) Write a c program for binary search and find the given no from list key= 21 5 9 13 16 19 21
9) Write a c code to insert and delete elements in circular queue.
10) Explain how to insert new node at beginning and delete node from middle in singly linked list.
11) Write a note on
    a) Fseck( ), ftell ( )
    b) Static memory
    c) Expand LIFO, FIFO
    d) POP ( )
    e) Priority queue

SEP 2018 DATA STURCTURES IN C 80 MARKS

SECTION -A

ANSWER ALL QUESTIONS. **(10*2=20)**

1)
   a) Define Data structures. Mention its types.
   b) Expand LIFO and FIFO
   c) Mention the applications of stack.
   d) Define linked list.
   e) List different tree traversal techniques.
   f) Define recursion.
   g) Define Height and level of a tree.
   h) Mention the constraints of “tower of Hanoi” problem while transferring a discs.
   i) What is infix & postfix expression.
   j) Define complete Binary tree.
   k) Write the difference between stack & queue.
   l) List the file I/O functions.

SECTION – B

Answer any four questions. **4*5=20**

2) Explain realloc( ) and malloc ( ) functions.
3) Define structure. Write its syntax. Create structure variable.
4) Explain various operations of queue.
5) Write ‘C’ program to find factorial of given number using recursion
6) Explain singly linked list.
7) Explain file error handling functions.

SECTION – C

Answer any four questions. **4*10=40**

8) Convert following infix expression to postfix expression.
   a) A+B-C*D
   b) A-B*C-D/E*F
   c) (A+B)*(D-C)
9)
   a) Write a C program to perform Arithmetic operations on pointers.
   b) Explain getw() and putw() functions.
10) Explain PUSH() & POP() AND DISPLAY() functions of stack.
11) Explain circular linked lists and doubly linked list.
12) Define following terms with an example.
   a) Root
   b) Siblings
   c) Degree of a node
   d) Path
   e) ancestors

SEP 2019 DATA STURCTURES IN C 80 MARKS

SECTION -A

ANSWER ALL QUESTIONS. **10*2=20**

1.
(a) What is pointer to pointer? How to declare it/
(b) What is a file? Mention its different access modes.
(c) What is static Memory Allocation?
(d) What are primitive and non-primitive data structures? Give examples.
(e) What is LIFO? List applications of it.
(f) What are the drawbacks of ordinary Queue?
(g) Write the syntax of any two file input functions.
(h) Differentiate between singly linked list and doubly linked list.
(i) What is empty linked list?
(j) What is priority queue? Mention different types of it.

SECTION – B

Answer any four questions. **4*5=20**

2. Write a program in c to print sum of array elements using Dynamic Memory Allocation.
3. Explain the concept of bubble sort technique with an example.
4. Write a note on Tower of Hanoi.
5. Define Stack. Explain various operations performed on stack.
6. Explain circular linked list.

SECTION – C

Answer any four questions. **4*10=40**

7. What is Dynamic Memory Allocation? Explain DMA functions.
8. (a) Write a program in C to print n Fibonacci numbers using recursion.
   (b) Convert the following expression to postfix :
      (i) ((A*(B+C))/(D)-F
      (II) (x+y)*(m/n+d)
9. Apply quicksort technique and sort the following numbers in ascending order. Also Write quicksort function for the same.
   56 , 24 , 43 , 92 , 52 , 41 , 32.
10. What is queue? Explain different types of queues.
11. Write Short notes on any two :
   (i) Pointer and structures.
   (ii) Components of linked list.
   (iii) File I/O functions.

SEP 2020 DATA STURCTURES IN C 80 MARKS

SECTION -A

ANSWER ALL QUESTIONS. **10*2=20**

1.
a) List advantages of pointer.
b) State the use of calloc () function with syntax.
c) State the purpose of rewind() function.
d) List applications of data structures.
e) What is recursion?
f) Evaluate the following postfix expression 12-34-*.
g) What is circular Queue?
h) Differentiate between linear search and binary search.
i) State advantages of linked list.
j) Define a node in a singly linked list.

SECTION – B

Answer any four questions. **4*5=20**

2. Write a program in C to read a text file and convert the file contents in uppercase and write the contents in an output file.
3. Discuss the classification of data structure with example.
4. Define Stack. Explain various operations performed on stack.
5. Explain double ended Queue.
6. Write a note on doubly linked list.

SECTION – C

Answer any four questions. **4*10=40**

7. What is file? Explain with syntax two input and two output functions related to file.
8.
   a) Explain the merge sort technique with example.
   b) Write a program in C to find GCD of two numbers using recursion.
9.
   a) Convert the following expression to postfix expression.
      i. ((A+(B-C)*D)^E+F)
      ii. (a+(b*c)-d)/e.
   b) Write a program in C to search an element in an array using sequential search.
10. Write a program in C to perform all operations on ordinary Queue.
11. Write short notes on any two:
    i. Dynamic Memory Allocation
    ii. Tower of Hanoi
    iii. Conversion of expression using Stack.

SEP 2021 DATA STURCTURES IN C 80 MARKS

SECTION -A

ANSWER ALL QUESTIONS. **10*2=20**

1.
   a) What is structure? Give example.
   b) Write a syntax of getc () and putc() functions.
   c) What are primitive and non primitive data structures?
   d) Convert the following expressions from infix to postfix:
      (A+B)*C/D.
   e) Define recursion.
   f) What is FIFO?
   g) What are the advantages of a Linked list?
   h) What are self referential structures?
   i) Define Binary tree.
   j) Which are the different traversal techinique for a binary tree?
   k) What is pointer?
   l) List the applications of a Queue.

SECTION – B

Answer any four questions. **4*5=20**

2. Write a program to implement the working of a Stack.
3. Differentiate between Static and dynamic allocation methods.
4. Write a program to demonstrate the working of a simple queue.
5. Write a note on representation of a Linked list.
6. Explain complete binary tree and binary search tree.
7. Write a note on error handling in files.

SECTION – C

Answer any four questions. **4*10=40**

8. Write a C program to create a file to store employee details like eno., ename and salary and display its contents in proper format.
9. Explain the classification of data structures in detail.
10.
   a) Write a program to find binomial coefficient using recursion.
   b) Explain the primitive operations on a stack.
11.
   a) Explain circular Queue and its operations using recursion.
   b) Define the following :
      i. Node
      ii. Degree of a node
      iii. Root
12. Explain the different types of linked list with diagram.

SEP 2022 DATA STURCTURES IN C 80 MARKS

SECTION -A

ANSWER ALL QUESTIONS. **10*2=20**

1.

a) What is data structures? Mention its Applications.
b) Define structure and union.
c) What is the purpose of getw() and putw() functions.
d) What is LIFOdata structures? How do you declare it using structure?
e) What is the purpose of fwrite() ststement?
f) What is the use of sizeof() and free() functions?
g) State advantages of circular queue.
h) What is double ended Queue? Mention its types.
i) What is self referential structure?
j) What is empty list?
k) Define strictly binary tree.
l) Define level of a tree.

SECTION – B

Answer any four questions. **4*5=20**

2. Explain file error handling functions.
3. Compare malloc() and calloc() functions.
4. Write a C program to print nth Fibonacci number using recursion.
5. Write following functions to demonstrate stack operations:
   i. Push()
   ii. Pop()
6. Differentiate between singly linked list and doubly linked list.
7. Explain complete binary tree and binary search tree.

SECTION – C

Answer any four questions. **4*10=40**

8. Write a program in C to create a file and display its content in proper format using following fields –Book_id, Book_name , Author and Price_of_book.
9. Convert the following infix expressions to postfix and prefix expressions.
   i. X*y-z^M+N-P/Q. ii. A*B-C+S/(E+F). iii. (AV)*(D-C).
10.
   a) Differentiate between ordinary queue and circular queue.
   b) Write a note on priority queue.
11. Explain the following terms.
   i. Ancestors ii. Forest
   ii. Forest iv. PathPath
   iii. Non-terminal node
   iv. Degree of a tree.
12. Write a short notes on any two
   i. Pointers
   ii. File I/O functions iii. Circular linked list.

SEP 2023 DATA STURCTURES IN C 80 MARKS

SECTION -A

ANSWER ALL QUESTIONS. **10*2=20**

1.

a) What is pointer? Write any two advantages.
b) Define Sequential search. List its advantages.
c) What are the various file positioning functions?
d) Explain realloc() function with syntax?
e) Differentiate between linear search and binary search.
f) Define non-Primitive data structures. Give example.
g) List out applications of stack.
h) Define the terms: (a) degree of node (b) circular queue.
i) What is linked list? How do you define node?
j) What is strictly binary tree? Give Example.
k) Define infix, prefix, postfix, expression.
l) What is the use of fscanf() and feof() fucntions?

SECTION – B

Answer any four questions. **4*5=20**

2. Write a program to implement tower of Hanoi problem.
3. Define stack. Explain stack operations with an example.
4. Write a program to sort an element using bubble sort.
5. Convert the following to postfix.
   i. (A+B)*(C+D)$(E+D)
   ii. (A+BCD)*(E+F/D)
6. What is circular queue? Explain how it is different from an ordinary queue.
7. Write a note on circular linked list.
8. Explian pre-order , post-order, for binary tree.
9. Explain the working of doubly linked lsit with neat diagram.

SECTION – C

Answer any four questions. **4*10=40**

10.
   a) Define file. Explain file handling functions.
   b) Write a note on dynamic memory allocation.
11. What is queue? Explain the operations of double ended queue.
12. Describe different insertion of nodes in linked list.
13. Write a program on quick sort, apply technique to sort the numbers 47, 60, 73, 54, 14, 33, 25.
14. Briefly explain (a) complete binary tree (b) binary search tree.

SEP 2023 DATA STURCTURES IN C 80 MARKS

SECTION -A

ANSWER ANY TEN QUESTIONS. **10*2=20**

1.

a) Define Union. Write the advantages of Union.
b) What is pointer? Which two operators are used to handle pointers?
c) What is the use of realloc() function?
d) What is the purpose of fseek() function?
e) List different types of data structures with examples.
f) Mention applications of stack.
g) What is priority queue? What are its different types?
h) State advantages of recursion.
i) Define (1) Null Pointer (2)External Pointer.
j) What is the different between singly linked list and doubly linked list?
k) What is complete binary tree?
l) How to represent a node in a binary tree? Give example.

SECTION =B **4*5=20**

Answer any four Questions.

2. Write a program using structures which consists of employee number, name of the employee and salary. Accept data of n employees and display it in proper format.
3. Explain with syntax two input and two output functions related to file.
4. Discuss conversion of Arithmetic expressions using Stack.
5. Write a program to find the binomial Co-effiecient using recursion.
6. Write a function LINSERT to insert a node in the front of a single linked list. Each node consists of roll no and name.
7. Explain traversal of a binary tree.

SECTION-C **4*10=40**

Answer any four Questions.

8. (a) Differentiate between static and dynamic memory allocation.
(b) Explain file error handling functions.
9. Write a program to demonstrate stack operations using recusion.
10. Write notes on any two:
(a) Circular queue (b) Double ended queue (c) Tower of Hanoi.
11. (a) Differentiate between circular linked list and circular doubly linked list.
(b) What are advantages and disadvantages of linked list?

12. (a) Define the following terms:

i. Degree of a tree ii. Sibiling iii. path

(b) Differentiate between binary tree and binary search tree.

SEP 2024 DATA STURCTURES IN C 80 MARKS

SECTION -A

ANSWER ANY TEN QUESTIONS. **10*2=20**

1. What is Data Structures?
2. Write a syntax of fread().
3. What is Node?
4. Write an algorithm for binary search method.
5. What is LIFO?
6. Define binary tree.
7. Compare single linked list and doubly linked list.
8. What is priority queue?
9. What is the use of getc() and putc() functions?
10. What is recursion?
11. What is complete binary search?
12. Write the differentiate between searching and sorting?

SECTION-B

Answer anty six questions? **4*5=20**

13. Explain the classification of data structure.
14. Explain briefly rewind(), fseek() and ftell() file positioning functions.
15. Write a program to find the factorial of n numbers using recursion.
16. Evaluate the following expressions into postfix expressions:
    (a) ((p+(q-r)*s)^t+u)
    (b) (A+(B-C)*D)
17. What is Linked list? Explain differentiate types of linked list.
18. Explain any 5 tree terminological associated with tree.
19. Write any 5 differentiate between iteration and recursion.
20. Write a C program to sort a list of N elements using Bubble sort technique.

SECTION-C

Answer any three Questions 4*10=40

21. What is dynamic memory allocation? Explain various memory management functions in C?
22. What is stack? Explain various operations that can be performed on stack.
23. What is queue? Explain briefly different types of queue.
24. Compare :
    (A) Bubble Sort and Selection Sort.
    (B) Linear search and binary search.
25. Write a short note on any two:
    (a) Merge sort
    (b) Infix and postfix expressions
    (c) Binary search
    (d) Static and dynamic memory allocation

SEP 2024 DATA STURCTURES IN C 80 MARKS

SECTION -A

ANSWER ANY TEN QUESTIONS. **10*2=20**

1. List the advatages of Pointer.
2. Mention the types of memory allocation techniques.
3. Define Datastructure and mention ite types.
4. List the different modes to open the file.
5. Define Recursion. Give an Example.
6. What is Stack? List its applications.
7. Mention the types of Queue.
8. Define height and level of Queue.
9. What is sorting? List the sorting techniques.
10. Defferentiate between linear search and binary search.
11. What is complete binary tree.
12. Mention the types of Traversal.

SECTION-B

Answer anty six questions? **4*5=20**

13. State the difference b/w static and dynamic memory allocation.
14. Explain the file error handling functions.
15. Write a Porgram in "C" to sort elements in ascending order using bubble sort.
16. Explain double ended queue.
17. Explian different typesof linked list with examples.

SECTION-C

Answer any three Questions 4*10=40

18. (a) Differentiate between malloc() and calloc().
    (b) What is Pointer? Write the advantages and disadvantages of Pointer.

19. (a) Write a "c" program to generate "n" Fibonacci numbers using recursive function.

    (b) What is file? Explain any five file I/O functions.

20. (a) Sort the following elements using selection sort method.

    15, 8, 7, 19, 1, 5, 7

    (b) Write "C" Program to search on element using Binary search technique.

21. (a) Convert the following infix expression to post fix and prefix form.

    $a \quad b \quad c - d + e + f/g.$

    (b)Write a "C" Program to demonstrate the working of simple queue.

22. (a) Define the following terms:

    i) Root ii) Node iii) Subtree iv) Sibling v) Path

    (b) Define Linked List. Explain its Components with neat diagram.

I HOPE ITS USEFULL FOR YOU